Lecture 2:
Truth Tables

NOT, AND, OR, THEN

P	¬ P
F	T
T	F

P	Q	P ∧ Q
F	F	F
F	T	F
T	F	F
T	T	T

P	Q	P ∨ Q
F	F	F
F	T	T
T	F	T
T	T	T

P	Q	P → Q
F	F	T
F	T	T
T	F	F
T	T	T

P V ¬ P

P	¬ P	P V ¬ P
F	T	T
T	F	T

$P \land \neg P$

P	$\neg P$	$P \land \neg P$
F	T	F
T	F	F

Equivalence

$\neg (P \vee Q) = \neg P \wedge \neg Q$

P	Q	P ∨ Q	¬ P	¬ Q	¬ (P ∨ Q)	¬ P ∧ ¬ Q	
F	F	F	T	T	T	T	
F	T	T	T	F	F	F	
T	F	T	F	T	F	F	
T	T	T	F	F	F	F	

Homework

$P \to (Q \to R) = (P \land Q) \to R$

P	Q	R						$P \to (Q \to R)$	$(P \land Q) \to R$
F	F	F							
F	F	T							
F	T	F							
F	T	T							
T	F	F							
T	F	T							
T	T	F							
T	T	T							